

02YMECH - Homework 11

Assigned for the week of Dec 1, 2025

Questions

1. Calculate the gravitational potential due to a thin rod of length l and mass m at a distance z from the center of the rod along its axis.

Hint: Use the integral

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

2. Calculate the gravitational potential due to a uniform ring of radius a and mass m at a point on the axis of the ring a distance z from its center.
3. Calculate the gravitational field inside and outside a spherical mass distribution of radius R and mass M with a density that varies with radius as

$$\rho(r) = \rho_0 \frac{r}{R}$$

4. A particle is dropped into a hole that goes through the center of the Earth. Neglecting rotational effects, and assuming the Earth has uniform density, show that the particle undergoes simple harmonic motion. What is the period of the oscillation in minutes? (Find the values of relevant constants on the internet as needed)